

Assignment 1

Research question

“This study examines how labor market conditions and inflation influence retail e-commerce sales in Canada. Using monthly seasonally adjusted data from Statistics Canada (2017–2025)”

Dataset and the variables

- Dependent variable: Monthly retail trade and e-commerce sales
- Independent Variable 1: Employment rate
- Independent variable: Employment, average hourly and weekly earnings (including overtime)
- Independent variable: Base Year: (2002 =100)

The data covers 8 years from Jan 2017 to Dec 2025 with monthly frequency all over Canada. I made sure all the extracted data are seasonally adjusted

- The dataset name: ecommerce.csv
- Number of observations: 107
- Number of variables: 5

df

A data.frame: 107 × 5				
date	ecommerce_sales	wage	employment_rate	cpi
<chr>	<int>	<dbl>	<dbl>	<dbl>
2017-01	1236885	967.02	61.4	130.1
2017-02	1212033	964.51	61.5	129.9
2017-03	1255537	966.94	61.6	129.7
2017-04	1281690	971.17	61.6	130.1
2017-05	1262875	970.08	61.6	129.9
2017-06	1276594	972.01	61.8	129.9

**1) Test the joint significance of your model at a 95% confidence level.
That is, calculate and interpret the F statistics.**

- Null hypothesis, where Employment, wages, and CPI have no joint effect on e-commerce sales.
- Alternative hypothesis: At least one coefficient is not equal zero

```
# Estimate linear regression model
model <- lm(ecommerce_sales ~ wage + employment_rate + cpi, data = df)
summary(model)
```

```
Call:
lm(formula = ecommerce_sales ~ wage + employment_rate + cpi,
    data = df)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-734010 -209414  -79258   222735  856915
```

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -4747374    1857011  -2.556   0.012 *
wage           19799       1971   10.044 < 2e-16 ***
employment_rate -7242     34270  -0.211   0.833
cpi           -96900     17744  -5.461  3.3e-07 ***
---

```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 335100 on 103 degrees of freedom
Multiple R-squared:  0.9138, Adjusted R-squared:  0.9113
F-statistic: 364.1 on 3 and 103 DF,  p-value: < 2.2e-16
```

- At level of confidence 95%, Alpha (1-0.95) = 0.05
- P value < 2.2e-16
- F-statistics: 364.1

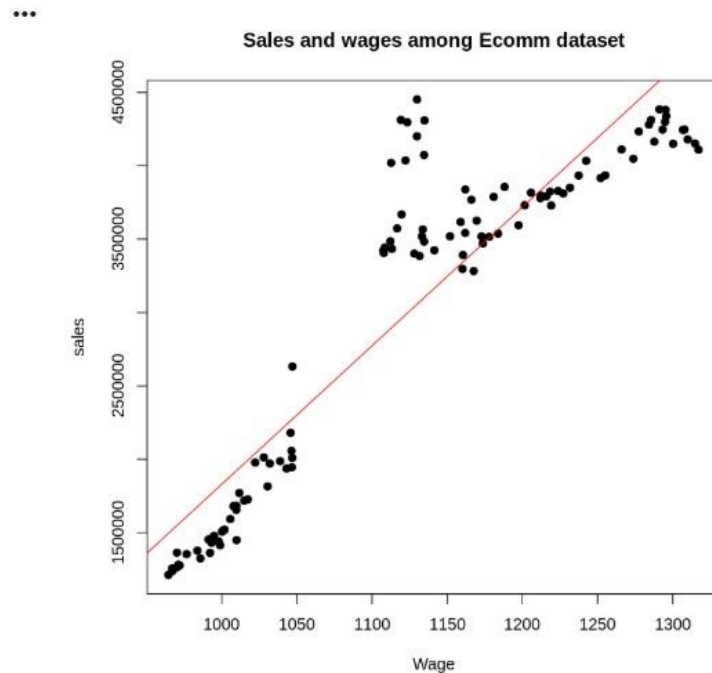
The overall P value associated with the F-statistics is less than 0.05 which means rejecting the null hypothesis (that all slope coefficients are jointly equal to zero). The independent variables of wages, employment rate and CPI have significant effect on the E-commerce sales. Economic conditions (wages, employment, and inflation) are strongly associated with online spending in Canada.

2) What can you say about heteroskedasticity in your model? Plot the residuals of your model and conduct the necessary test to identify the presence of this problem.

We usually assume that the error term is constant during OLS estimation. That is, we assume there's a constant amount of "randomness" in the real world that affects E-commerce sales overtime.

We're going to observe the spread or distribution of the error term (which captures the gap between the actual observation and its predicted value).

```
plot(df$wage, df$ecommerce_sales, main="Sales and wages among Ecomm dataset",  
     xlab="Wage", ylab="sales", pch=19)  
abline(lm(df$ecommerce_sales~df$wage), col="red") # regression line (y~x)
```



- The error (represented by the visual gap between the points and the regression line) looks consistently close to the line for most values up to income of 1050 a week, then it gets dispersed randomly from the regression line between 1100 to 1200 and finally gets back close to the line between 1200 and 1300.
- Since we can't spot an obvious visual pattern, we'll need to do a formal test for Heteroskedasticity.

Bruesh-Pagan Method

- Null (H0): The residuals are distributed with equal variance (homoscedastic)
- Alternative (HA): The residuals are distributed with unequal variance (heteroskedastic)

```
library(lmtest)
lm.Data2 = lm(ecommerce_sales ~ wage + employment_rate + cpi, data = df)
summary(lm.Data2)
bptest(lm.Data2)
```

```
*** Loading required package: zoo
```

```
Attaching package: 'zoo'
```

```
The following objects are masked from 'package:base':
```

```
as.Date, as.Date.numeric
```

```
Call:
```

```
lm(formula = ecommerce_sales ~ wage + employment_rate + cpi,
    data = df)
```

```
Residuals:
```

```
      Min       1Q   Median       3Q      Max
-734010 -209414  -79258   222735  856915
```

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-4747374	1857011	-2.556	0.012 *
wage	19799	1971	10.044	< 2e-16 ***
employment_rate	-7242	34270	-0.211	0.833
cpi	-96900	17744	-5.461	3.3e-07 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 335100 on 103 degrees of freedom
```

```
Multiple R-squared:  0.9138, Adjusted R-squared:  0.9113
```

```
F-statistic: 364.1 on 3 and 103 DF,  p-value: < 2.2e-16
```

```
studentized Breusch-Pagan test
```

```
data: lm.Data2
```

```
BP = 13.853, df = 3, p-value = 0.003112
```

The P-value of $0.003112 < 0.05$, so we reject the null hypothesis, and the model has unequal variance. As we have heteroskedastic problem, we will dig further in detail using the Robust standard error.

3) Report the Robust standard error of your model. Do you think this has fixed any heteroskedasticity problem in your model?

Due to the presence of heteroskedasticity, we're going to adjust the standard errors of coefficient estimates using the feasible generalized least square (FGLS) makes the residuals of a model more like one another (reduces the variance of the errors).

We create the regression model and generate the model summary, so we can compare the original standard errors with the robust standard errors later.

```
library(lmtest)
lm.Data3 = lm(formula = ecommerce_sales ~ wage + employment_rate + cpi, data = df)
summary(lm.Data3)
```

...

Call:
lm(formula = ecommerce_sales ~ wage + employment_rate + cpi,
data = df)

Residuals:

	Min	1Q	Median	3Q	Max
	-734010	-209414	-79258	222735	856915

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-4747374	1857011	-2.556	0.012 *
wage	19799	1971	10.044	< 2e-16 ***
employment_rate	-7242	34270	-0.211	0.833
cpi	-96900	17744	-5.461	3.3e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 335100 on 103 degrees of freedom
Multiple R-squared: 0.9138, Adjusted R-squared: 0.9113
F-statistic: 364.1 on 3 and 103 DF, p-value: < 2.2e-16

Run `coeftest` to correct the `lm.Data3` model's standard errors (`vcov`) by making them robust using the `HC0` method:

```
library(sandwich)
coeftest(lm.Data3, vcov = vcovHC(lm.Data3, type = "HC0"))
```

The regression new output with the robust Se

```
#This is regression with robust se
lm.Data3 = lm(ecommerce_sales ~ wage + employment_rate + cpi, data = df)
summary(lm.Data3)
coeftest(lm.Data3, vcov = vcovHC(lm.Data3, type = "HC0"))
```

```
***
Call:
lm(formula = ecommerce_sales ~ wage + employment_rate + cpi,
    data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-734010 -209414  -79258  222735  856915

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -4747374    1857011  -2.556   0.012 *
wage         19799      1971    10.044 < 2e-16 ***
employment_rate -7242    34270  -0.211   0.833
cpi         -96900     17744   -5.461  3.3e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 335100 on 103 degrees of freedom
Multiple R-squared:  0.9138, Adjusted R-squared:  0.9113
F-statistic: 364.1 on 3 and 103 DF,  p-value: < 2.2e-16

t test of coefficients:

            Estimate Std. Error t value Pr(>|t|)
(Intercept) -4747373.7  1390391.3  -3.4144 0.0009154 ***
wage         19799.0    2129.0    9.2998 2.666e-15 ***
employment_rate -7241.5  27920.8  -0.2594 0.7958771
cpi         -96899.6    19517.0  -4.9649 2.734e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Now, we can compare the main model and the model with robust se. The coefficient estimates remain unchanged, but the standard errors differ slightly from the original OLS estimates as follows:

The standard error for the wage variable increased from 1,971 to 2,129, while the CPI standard error increased from 17,744 to 19,517. These changes slightly reduce the t-statistics but do not alter the overall inference. wages and CPI remain statistically significant determinants of e-commerce sales, while the employment rate remains insignificant.

**4) What about autocorrelation? Do you think your model suffers from this?
Conduct the necessary tests.**

OLS estimation assumes that error is constant, so if the correlation is not zero, we have autocorrelation.

```
# Run the regression model
model <- lm(ecommerce_sales ~ wage + employment_rate + cpi, data = df)

# View summary
summary(model)
```

Call:

```
lm(formula = ecommerce_sales ~ wage + employment_rate + cpi,
    data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-734010	-209414	-79258	222735	856915

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-4747374	1857011	-2.556	0.012 *
wage	19799	1971	10.044	< 2e-16 ***
employment_rate	-7242	34270	-0.211	0.833
cpi	-96900	17744	-5.461	3.3e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 335100 on 103 degrees of freedom

Multiple R-squared: 0.9138, Adjusted R-squared: 0.9113

F-statistic: 364.1 on 3 and 103 DF, p-value: < 2.2e-16

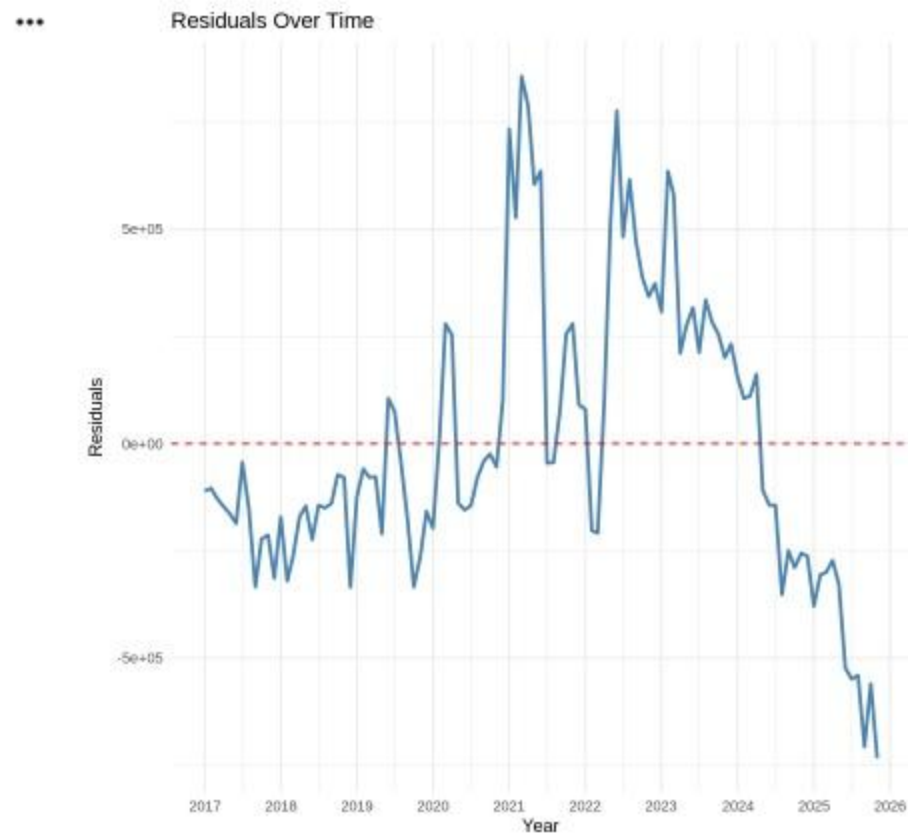
```
# Extract residuals
df$residuals <- resid(model)
```



```

ggplot(df, aes(x = date, y = residuals)) +
  geom_line(color = "steelblue", linewidth = 1) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "red") +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y") +
  labs(
    title = "Residuals Over Time",
    x = "Year",
    y = "Residuals"
  ) +
  theme_minimal()

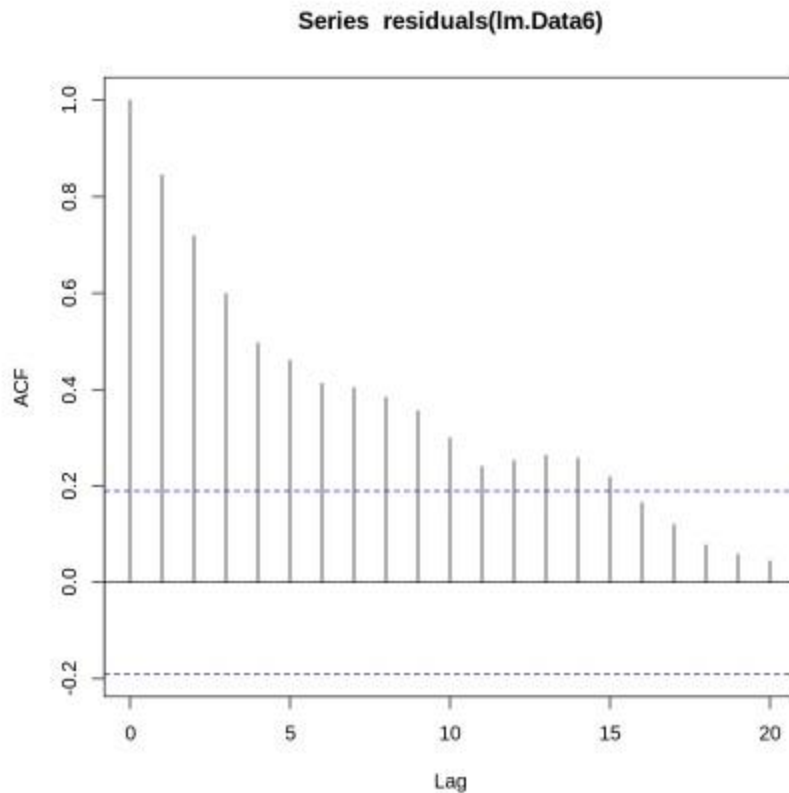
```



The data is spiky and random from 2017 to 2021, then exhibits a sharp upward trend from 2020 to the first half of 2021, followed by a large drop, then a sharp rise in 2022, after which it moves downward smoothly and is negatively autocorrelated. Since it's not clear enough to decide

Autocorrelation Function Let's use the acf function in R to see the correlation between residuals and their lags.


```
acf(residuals(lm.Data6))
```



The ACF plot shows that autocorrelation decreases as the lag increases, meaning observations that are closer in time are more correlated than those further apart. This suggests that the effect of past values gradually diminishes over time. Since this pattern appears in the residuals, it indicates that some temporal dependence may still exist and the model may not fully capture the time-series structure.

Durbin-Watson Test for Autocorrelation

- Null: no correlation among a model's residuals
- Alternative: the model's residuals are autocorrelated

```
lm.Datatest = lm(ecommerce_sales ~ wage + employment_rate + cpi, data = df)
summary(lm.Datatest)
```

Call:

```
lm(formula = ecommerce_sales ~ wage + employment_rate + cpi,
    data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-734010	-209414	-79258	222735	856915

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-4747374	1857011	-2.556	0.012 *
wage	19799	1971	10.044	< 2e-16 ***
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cpi	-96900	17744	-5.461	3.3e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 335100 on 103 degrees of freedom

Multiple R-squared: 0.9138, Adjusted R-squared: 0.9113

F-statistic: 364.1 on 3 and 103 DF, p-value: < 2.2e-16

```
dwtest(lm.Datatest)
```

Durbin-Watson test

data: lm.Datatest

DW = 0.26228, p-value < 2.2e-16

alternative hypothesis: true autocorrelation is greater than 0

As the p-value is much smaller than 0.05, DW = 0.262, which is very close to 0, indicates strong positive autocorrelation in the residuals.

At the 95% confidence level ($\alpha = 0.05$), we reject the null hypothesis of no autocorrelation. meaning the errors are correlated over time. Therefore, the regression model violates the independence assumption of the error terms, suggesting that the model may be mis-specified

5) Estimate a Log-Log model and interpret the coefficients.

```
lm.logData1 = lm(log(ecommerce_sales) ~ log(wage) + log(employment_rate) + log(cpi), data = df)
summary(lm.logData1)
```

Call:

```
lm(formula = log(ecommerce_sales) ~ log(wage) + log(employment_rate) +
    log(cpi), data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.273421	-0.100092	-0.002046	0.102110	0.290042

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-22.9822	4.0887	-5.621	1.63e-07	***
log(wage)	9.4295	0.8281	11.387	< 2e-16	***
log(employment_rate)	0.7651	0.7680	0.996	0.322	
log(cpi)	-6.3398	0.9712	-6.528	2.56e-09	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1311 on 103 degrees of freedom

Multiple R-squared: 0.9154, Adjusted R-squared: 0.9129

F-statistic: 371.5 on 3 and 103 DF, p-value: < 2.2e-16

In a log–log regression model, the coefficients represent elasticities, meaning they measure the percentage change in the dependent variable resulting from a 1% change in the independent variable, holding other factors constant.

- Wage has a strong positive and statistically significant effect on e-commerce sales, a 1% increase in weekly earnings is associated with a 9.4% increase in e-commerce sales.
- Employment rate does not have a statistically significant effect on e-commerce sales, a 1% increase in employment rate leads to about 0.77% increase in e-commerce sales.
- CPI is highly statistically significant, and a 1% increase in CPI leads to approximately a 6.34% decrease in e-commerce sales

6) Add fixed effects to your model. Interpret.

Year fixed effects were included to control unobserved time-specific shocks between 2017 and 2025 that may affect e-commerce sales. This ensures that the estimated relationships between independent variables and e-commerce sales are not biased by macroeconomic changes occurring in specific years.

```
fe_time_model <- lm(ecommerce_sales ~ (wage) +
  (employment_rate) +
  (cpi) +
  factor(year),
  data = df)

summary(fe_time_model)
```

Call:
lm(formula = (ecommerce_sales) ~ (wage) + (employment_rate) +
 (cpi) + factor(year), data = df)

Residuals:

Min	1Q	Median	3Q	Max
-440198	-79623	-14631	85940	432770

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3660967	1881893	1.945	0.054687 .
wage	11138	1575	7.074	2.55e-10 ***
employment_rate	-90958	20650	-4.405	2.78e-05 ***
cpi	-58280	16452	-3.542	0.000617 ***
factor(year)2018	113886	78862	1.444	0.151992
factor(year)2019	344259	100883	3.412	0.000947 ***
factor(year)2020	662933	151952	4.363	3.26e-05 ***
factor(year)2021	1377576	194029	7.100	2.25e-10 ***
factor(year)2022	1349587	277996	4.855	4.73e-06 ***
factor(year)2023	1492205	346610	4.305	4.06e-05 ***
factor(year)2024	1321476	404062	3.270	0.001497 **
factor(year)2025	1153294	451375	2.555	0.012205 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 170300 on 95 degrees of freedom
Multiple R-squared: 0.9795, Adjusted R-squared: 0.9771
F-statistic: 412.2 on 11 and 95 DF, p-value: < 2.2e-16

* Factor(year) automatically creates year dummy variables.

- Wage has a positive and statistically significant effect on e-commerce sales. A one-unit increase in wage is associated with an increase of approximately \$11,138 in e-commerce sales, holding other variables constant.

- Employment Rate has a negative and statistically significant effect on e-commerce sales. A one-unit increase in employment rate is associated with a decrease of approximately \$90,958 in e-commerce sales, holding other variables constant.
- CPI has a negative and statistically significant effect on e-commerce sales. A one-unit increase in CPI is associated with a decrease of approximately \$58,280 in e-commerce sales, holding other variables constant.

The coefficients for the years variables measure the difference in e-commerce sales relative to the base year 2017 which is omitted from the dataset. Most of the years are positive and statistically significant starting from 2019 to 2025. Also, we can see a sudden sales growth in the coefficients of years around the pandemic, for example in 2021, the sales increased by 1.3 billion than in 2017.

References

Course materials, personal notes

[\(51\) Math and Science - YouTube](#)

<https://www.datacamp.com/>

<https://www150.statcan.gc.ca>